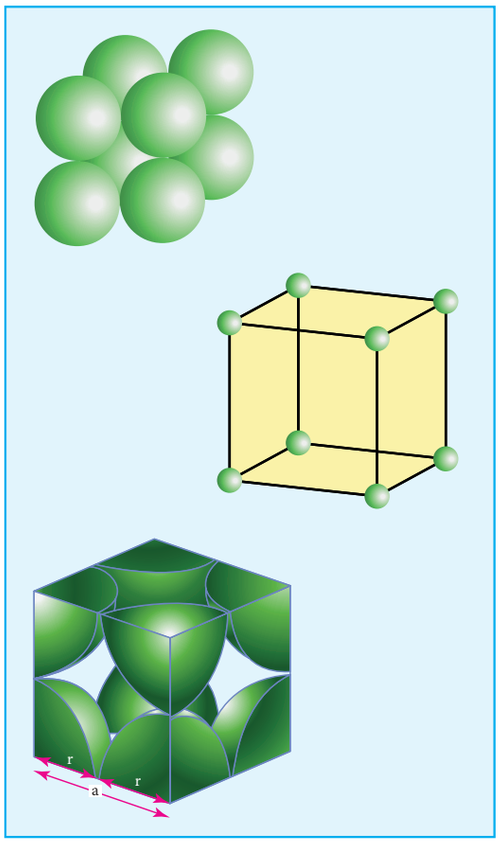
Number of atoms in a cubic unit cell:

6.5.1 Primitive (or) simple cubic unit cell.(SC)



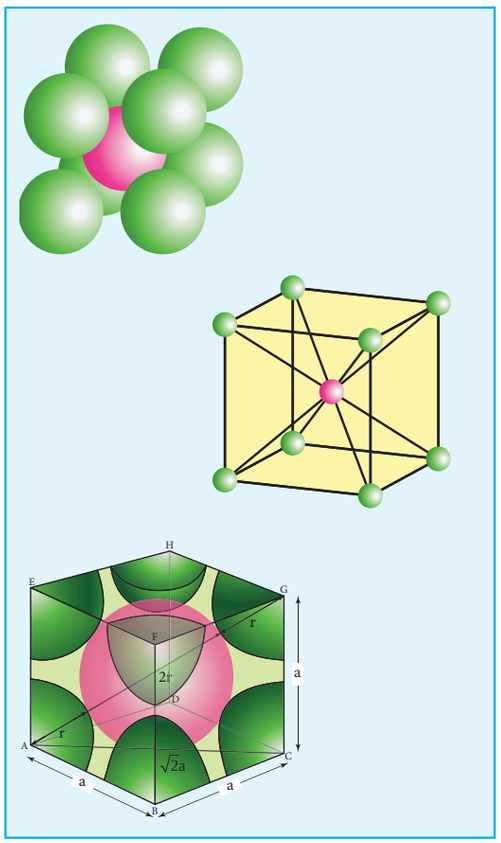
In the simple cubic unit cell, each corner is occupied by an identical atoms or ions or molecules. And they touch along the edges of the cube, do not touch diagonally. The coordination number of each atom is 6.

Each atom in the corner of the cubic unit cell is shared by 8 neighboring unit cells and therefore atoms `per unit cell is equal to $\frac{N_c}{8}$, where N_c is the number of atoms at the corners.

$$\therefore \text{no of atoms in a SC unit cell} = \left(\frac{Nc}{8}\right)$$

$$= \left(\frac{8}{8}\right) = 1$$

6.5.2 Body centered cubic unit cell. (BCC)



In a body centered cubic unit cell, each corner is occupied by an identical particle and in addition to that one atom occupies the body centre. Those atoms which occupy the corners do not touch each other, however they all touch the one that occupies the body centre. Hence, each atom is surrounded by eight nearest neighbours and coordination number is 8. An atom present at the body centre belongs to only to a particular unit cell i.e unshared by other unit cell.

$$\therefore \text{Number of atoms in a bccunitcell} = \left(\frac{Nc}{8}\right) + \left(\frac{N_b}{1}\right)$$

$$= \left(\frac{8}{8} + \frac{1}{1}\right)$$

$$= (1+1)$$

$$= 2$$

XII U6 Solid State - Jerald.indd 183 3/29/2019 6:30:19 PM